

KnowSelf

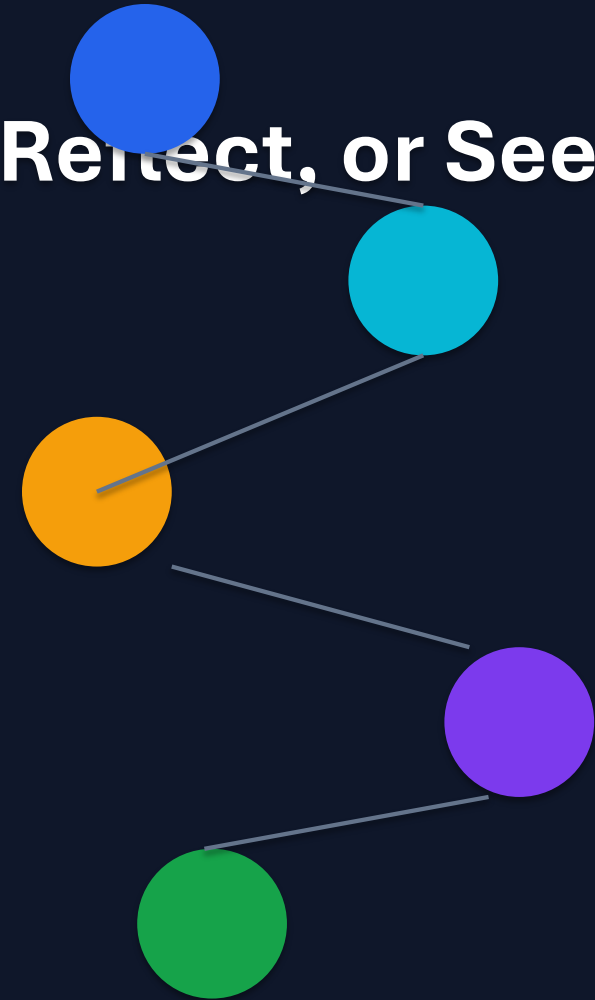
Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness for planning systems

FAST

SLOW

KNOWLEDGEABLE



Current agent training uses “flood irrigation”

Indiscriminate injection of trajectories, feedback, or domain knowledge ignores situational need.

Trajectory imitation (ReAct-style)

Gold trajectories shape behavior, but agents can collapse into brittle pattern imitation on unexpected inputs.

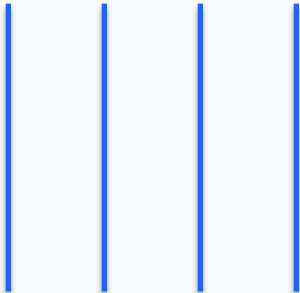
Trial-and-error refinement (Reflexion-style)

External feedback can improve traces, but reflection is applied without a criterion for when it is necessary.

Knowledge-augmented planning (KnowAgent / WKM)

Domain knowledge is injected broadly, increasing inference cost and sometimes hurting task success.

Flood irrigation



Every step receives the same resource

Planning failure modes

Fragile planning under distribution shift
Pattern collapse on unexpected states
Unnecessarily high inference costs

Human-like self-awareness

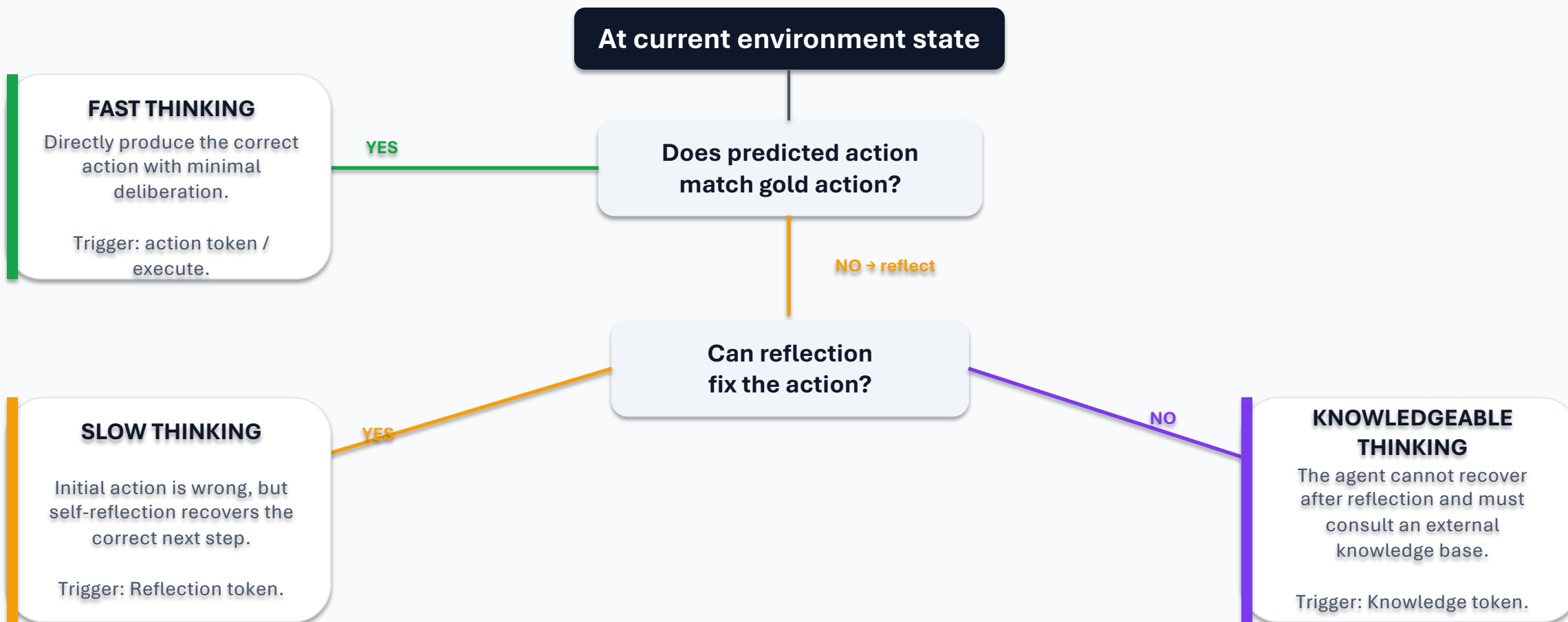
act directly

pause + reflect

seek external help

Three situational modes: fast, slow, knowledgeable

KnowSelf labels each decision step by what the agent can do before and after reflection.



Decision criterion is heuristic during data construction; learned special-token generation controls behavior at inference.

KnowSelf framework: data construction → training → inference

A lightweight pipeline converts self-explored trajectories into situational self-awareness.

1 Data construction

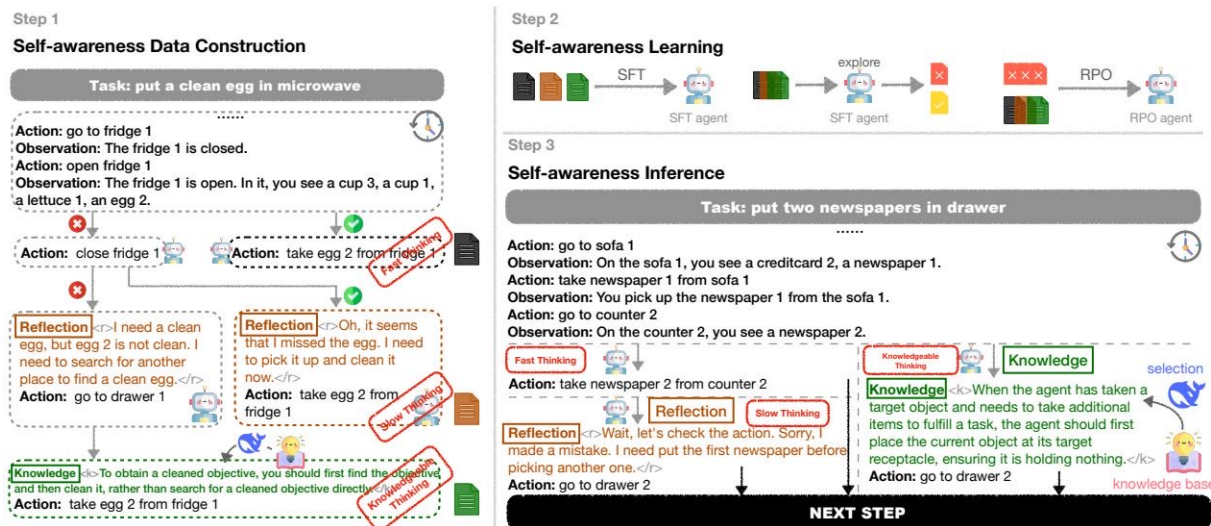
Self-explore environments; compare predicted action with gold action; test whether reflection fixes the step; insert special tokens.

2 Two-stage training

SFT teaches initial self-awareness patterns;
RPO refines the policy for generating the appropriate special token.

3 Self-aware inference

At deployment, generated tokens route the agent to direct action, reflection, or external knowledge retrieval.



Framework figure from source brief: construction, learning, and inference are linked by special situation tokens.

Qualitative examples: KnowSelf recovers where strong reasoners fail

Two ALFWorld cases illustrate selective reflection and knowledge retrieval.

(a) Task: put a hot mug in cabinet	
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge]<k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection]<r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge]<k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

Case A: “put a hot mug in cabinet”

OpenAI-O1 opens the microwave but then incorrectly tries to take an egg. KnowSelf emits a Reflection token and realizes the mug should be heated using the open microwave.

Case B: “put a saltshaker in drawer”

DeepSeek-R1 gets stuck trying to close an open drawer first. KnowSelf emits a Knowledge token: place the object without removing unrelated items.

Interpretation: the agent does not always need more context; it needs a learned decision about which cognitive resource to invoke.

KnowSelf reaches 84.33% on ALFWorld with only 15.01% knowledge

Success rates (%) across six ALFWorld task types; Know% is the proportion of steps using external knowledge.

Llama-8B KnowSelf

84.33% All

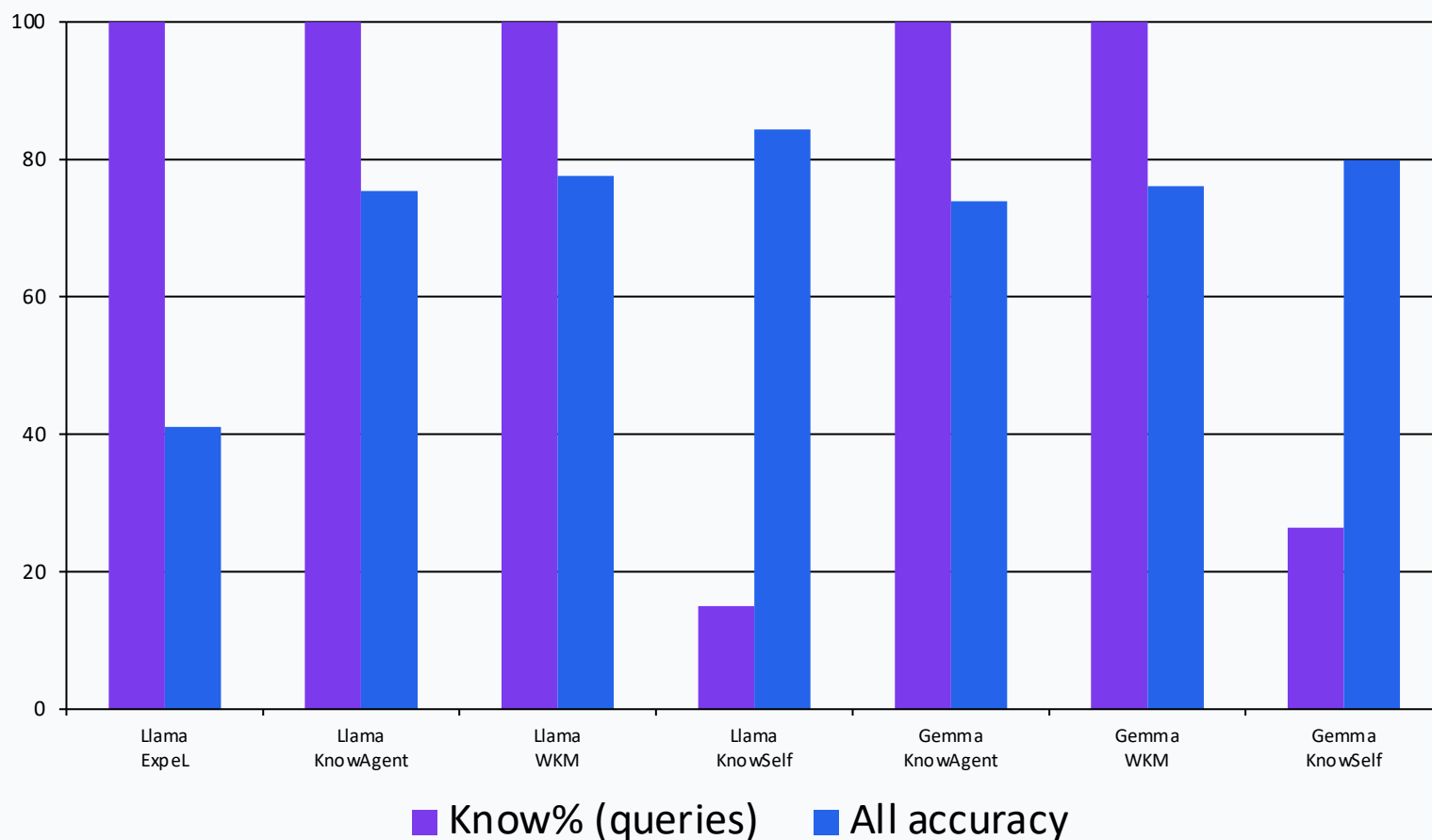
15.01% Know

Best Llama-8B overall while using far less knowledge than 100% methods.

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Minimal knowledge, maximum impact

KnowSelf reduces knowledge calls to ~15–26% while retaining or improving overall ALFWorld accuracy.



Efficiency readout

Full-knowledge baselines query knowledge at every step (100%).
KnowSelf queries selectively: 15.01% for Llama-8B and 26.41% for Gemma-2B.

Scaling behavior

Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpL, while Llama-8B KnowSelf reaches 84.33% overall.

Core implication

The bottleneck is not simply access to more knowledge; it is choosing when knowledge is warranted.

Key takeaways

Self-awareness is the missing control policy for agent planning.

1

Indiscriminate knowledge injection is wasteful and can be suboptimal.

2

Special tokens let agents learn situational self-awareness from lightweight heuristic labels.

3

Llama-8B KnowSelf outperforms strong baselines, including GPT-4o ReAct (84.33% vs 76.12%) and GPT-4o ExpeL (79.85%).

4

Only 15.01–26.41% knowledge usage is needed versus 100% for competing knowledge-heavy methods.

5

The paradigm scales down: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL.

Future directions

Broader environments · richer retrieval policies · calibration of when reflection is enough · cost-aware planning objectives